

Hypothesis Testing

Question no 1:- What is a null hypothesis (H_0) and why is it important in hypothesis testing?

Answer:- The null hypothesis (H_0) is a fundamental concept in statistics that represents the default or baseline assumption in hypothesis testing.

Definition

The null hypothesis states that there is no significant effect, difference, or relationship between variables in the population.

It's the statement we test against to determine whether the observed data provide enough evidence to support an alternative hypothesis (H_1).

For example:

In testing a new drug:

- H_0 : The drug has no effect on recovery time.

In comparing two teaching methods:

- H_0 : There is no difference in average test scores between methods.

Importance in Hypothesis Testing

Provides a baseline for comparison

1. It defines what we expect if no real effect exists, allowing us to measure deviations objectively.

Enables statistical decision-making

2. By calculating a *p-value* or test statistic, we can decide whether to reject H_0 in favor of H_1 .

Controls error rates

3. It helps manage the probability of making a **Type I error** (rejecting a true H_0) and ensures scientific rigor.

Supports objectivity

4. Researchers start with the assumption that nothing unusual is happening—reducing bias and promoting evidence-based conclusions.

In short, the null hypothesis is the **anchor of statistical inference**—it ensures that conclusions are drawn from data, not assumptions.

Hypothesis Testing

Question no 2:- What does the significance level (α) represent in hypothesis testing?

Answer:- The significance level (α) in hypothesis testing is the threshold we set to decide whether to reject the null hypothesis (H_0).

Definition

- It represents the probability of making a Type I error — rejecting H_0 when it is actually true.
- Common choices are $\alpha = 0.05$ (5%), $\alpha = 0.01$ (1%), or $\alpha = 0.10$ (10%), depending on how strict we want to be.

It Matters

Controls false positives

1. By setting α , we limit the chance of wrongly concluding that an effect exists when it doesn't.
 - Example: With $\alpha = 0.05$, we accept a 5% risk of incorrectly rejecting H_0 .

Defines the decision boundary

If the $p\text{-value} \leq \alpha \rightarrow$ reject H_0 (evidence supports H_1).

2. If the $p\text{-value} > \alpha \rightarrow$ fail to reject H_0 (insufficient evidence).
3. **Balances risk and sensitivity**
 - A **smaller α** (like 0.01) reduces false positives but makes it harder to detect real effects.
 - A **larger α** (like 0.10) increases sensitivity but risks more false positives.

Example (Retail Dataset Context)

Suppose we test whether **average sales differ between East and West regions**:

- $H_0: \mu_{\text{East}} = \mu_{\text{West}}$
- $H_1: \mu_{\text{East}} \neq \mu_{\text{West}}$

If we set $\alpha = 0.05$, we only reject H_0 if the probability of observing our data under H_0 is less than 5%. This ensures we're cautious about claiming a difference unless the evidence is strong.

In short, α is the “**risk tolerance**” for false alarms in hypothesis testing.

Question no 3:- Differentiate between Type I and Type II errors.

Answer:- These two errors are at the heart of hypothesis testing.

Hypothesis Testing

Type I Error (False Positive)

- Occurs when we reject the null hypothesis (H_0) even though it is actually true.
- In other words, we conclude there is an effect or difference when none exists.
- Probability of committing this error = significance level (α).
- Example: A retailer believes a new marketing campaign increases sales. If H_0 (no effect) is true but we reject it, we wrongly assume the campaign works.

Type II Error (False Negative)

- Occurs when we fail to reject the null hypothesis (H_0) even though it is false.
- In other words, we miss a real effect or difference.
- Probability of committing this error = β .
- The complement ($1 - \beta$) is called statistical power, which measures the ability to detect true effects.
- Example: If the campaign truly increases sales but we fail to reject H_0 , we incorrectly conclude it has no impact.

Key Difference

- **Type I Error (α):** Detecting a difference that doesn't exist → false alarm.
- **Type II Error (β):** Missing a difference that does exist → missed opportunity.

Retail Dataset Analogy

Suppose we test whether **average revenue differs between East and West regions**:

- **Type I Error:** Concluding East $\neq$ West when in reality they are equal.
- **Type II Error:** Concluding East = West when in reality they differ.

In short:

- **Type I = false alarm** (too eager to reject H_0).
- **Type II = missed detection** (too cautious to reject H_0).

Question no 4:- Explain the difference between a one-tailed and two-tailed test. Give an example of each.

Answer:- A very important distinction in hypothesis testing is between one-tailed and two-tailed tests.

♦ One-Tailed Test

- Definition: Tests for an effect in one specific direction (greater than or less than).
- Use Case: When you have a clear directional hypothesis.

Hypothesis Testing

- Example:
 - Retail context: Testing if average sales in the East region are greater than in the West region.
 - $H_0: \mu_{\text{East}} \leq \mu_{\text{West}}$
 - $H_1: \mu_{\text{East}} > \mu_{\text{West}}$
 - Here, we only care if East > West, not if East < West.

Two-Tailed Test

- **Definition:** Tests for an effect in **either direction** (different, not equal).
- **Use Case:** When you want to check for **any difference**, regardless of direction.
- **Example:**
 - Retail context: Testing if **average sales in the East region differ from the West region**.
 - $H_0: \mu_{\text{East}} = \mu_{\text{West}}$
 - $H_1: \mu_{\text{East}} \neq \mu_{\text{West}}$
 - Here, we care about both possibilities: East > West or East < West.

Key Difference

- **One-tailed** → More powerful if you're only interested in one direction, but risks missing effects in the opposite direction.
- **Two-tailed** → More conservative, tests both directions, widely used in research when no strong directional expectation exists.

In short:

- One-tailed = directional (greater/less)
- Two-tailed = non-directional (different)

Question no 5:- A company claims that the average time to resolve a customer complaint is 10 minutes. A random sample of 9 complaints gives an average time of 12 minutes and a standard deviation of 3 minutes. At $\alpha = 0.05$, test the claim.

Answer:- Problem Setup

- Claimed mean resolution time: $\mu_0 = 10$ minutes
- Sample mean: $\bar{x} = 12$ minutes
- Sample size: $n = 9$
- Sample standard deviation: $s = 3$ minutes
- Significance level: $\alpha = 0.05$

Step 1: State Hypotheses

Hypothesis Testing

- Null Hypothesis (H_0): $\mu=10$ (average resolution time is 10 minutes)

Alternative Hypothesis (H_1): $\mu \neq 10$ (average resolution time is not 10 minutes)

- $\rightarrow$ This is a two-tailed test.

Step 2: Test Statistic

We use the **t-test** since the sample size is small and population σ is unknown:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$
$$t = \frac{12-10}{3/\sqrt{9}} = \frac{2}{1} = 2.0$$

Step 3: Degrees of Freedom

$$df = n - 1 = 9 - 1 = 8$$

Step 4: Critical Value

For a two-tailed test at $\alpha=0.05$ and $df=8$:

$$T_{\text{critical}} \approx \pm 2.306$$

Step 5: Decision

- Calculated $t=2.0$
- Critical $t=\pm 2.306$

Since $|t|=2.0 < 2.306$, we **fail to reject H_0** .

Conclusion

At the 5% significance level, the sample does **not provide enough evidence** to reject the company's claim.

We cannot conclude that the average resolution time is different from 10 minutes.

Question no 6:- When should you use a Z-test instead of a t-test?

Answer:- The choice between a Z-test and a t-test depends mainly on sample size and whether the population standard deviation (σ) is known.

Hypothesis Testing

♦ Use a Z-test when:

1. Population standard deviation (σ) is known.
 - Example: A company has historical data showing $\sigma = 5$ minutes for complaint resolution times.
2. Sample size is large ($n \geq 30$).
 - With large samples, the Central Limit Theorem ensures the sampling distribution of the mean is approximately normal.
3. Data are approximately normally distributed.

Use a t-test when:

1. **Population standard deviation (σ) is unknown.**
 - You estimate variability using the sample standard deviation (s).
2. **Sample size is small ($n < 30$).**
 - The t-distribution accounts for extra uncertainty in small samples.
3. **Data are approximately normally distributed.**

Example

Z-test example:

- A factory claims the average weight of a packaged product is 500g. The population $\sigma = 10$ g is known. A sample of 50 packages is tested. → Use a **Z-test**.

t-test example:

- A company claims the average resolution time is 10 minutes. A sample of 9 complaints has mean = 12 minutes, $s = 3$ minutes, and σ is unknown. → Use a **t-test** (like the problem you asked earlier).

In short:

- **Z-test** → known σ , large n .
- **t-test** → unknown σ , small n .

Question no 7:- The productivity of 6 employees was measured before and after a training program.

Employee	Before	After
1	50	55

Hypothesis Testing

2	60	65
3	58	59
4	55	58
5	62	63
6	56	59

At $\alpha = 0.05$, test if the training improved productivity.

Answer:- The training program improved productivity using a paired t-test (since the same employees were measured before and after).

Data

Employee	Before	After	Difference (After – Before)
1	50	55	+5
2	60	65	+5
3	58	59	+1
4	55	58	+3
5	62	63	+1
6	56	59	+3

Step 1: Hypotheses

- **H₀ (Null):** Training has no effect $\rightarrow$ mean difference = 0

H₁ (Alternative): Training improves productivity $\rightarrow$ mean difference > 0

- $\rightarrow$ This is a **one-tailed paired t-test**.

Step 2: Calculate Differences

- Differences = [5, 5, 1, 3, 1, 3]
- Mean difference ($\bar{d}$) =
$$\frac{5+5+1+3+1+3}{6} = \frac{18}{6} = 3$$
- Standard deviation of differences (s_d) ≈ 1.788

Hypothesis Testing

Step 3: Test Statistic

$$t = \frac{d^-}{\frac{sd/\sqrt{n}}{3}} = \frac{1.788/\sqrt{6}}{0.730} \approx 4.11$$

Step 4: Degrees of Freedom

$$df = n - 1 = 6 - 1 = 5$$

Step 5: Critical Value

At $\alpha = 0.05$ (one-tailed) and $df = 5$:

$$t_{\text{critical}} \approx 2.015$$

Step 6: Decision

- Calculated $t = 4.11$

$$\text{Critical } t = 2.015$$

- $\rightarrow$ Since $t > t_{\text{critical}}$, we **reject** H_0 .

Conclusion

At the 5% significance level, there is **strong evidence that the training program improved employee productivity**.

Question no 8:- A company wants to test if product preference is independent of gender.

Gender	Product A	Product B	Total
Male	30	20	50
Female	10	40	50
Total	40	60	100

Hypothesis Testing

At $\alpha = 0.05$, test independence

Answer:- The Chi-Square Test of Independence to check if product preference is independent of gender.

Step 1: Observed Frequencies

Gender	Product A	Product B	Total
Male	30	20	50
Female	10	40	50
Total	40	60	100

Step 2: Expected Frequencies

Formula:

$$E_{ij} = \frac{(\text{Row Total})(\text{Column Total})}{\text{Grand Total}}$$

- Male, Product A: $\frac{50 \cdot 40}{100} = 20$
- Male, Product B: $\frac{50 \cdot 60}{100} = 30$
- Female, Product A: $\frac{50 \cdot 40}{100} = 20$
- Female, Product B: $\frac{50 \cdot 60}{100} = 30$

Gender	Product A	Product B	Total
Male	20	30	50

Hypothesis Testing

Female	20	30	50
Total	40	60	100

Step 3: Chi-Square Statistic

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

- Male, A: $\frac{(30-20)^2}{20} = \frac{100}{20} = 5$
 - Male, B: $\frac{(20-30)^2}{30} = \frac{100}{30} \approx 3.33$
 - Female, A: $\frac{(10-20)^2}{20} = \frac{100}{20} = 5$
 - Female, B: $\frac{(40-30)^2}{30} = \frac{100}{30} \approx 3.33$
- $\chi^2 = 5 + 3.33 + 5 + 3.33 = 16.66$

Step 4: Degrees of Freedom

$$df = (rows-1)(columns-1) = (2-1)(2-1) = 1$$

Step 5: Critical Value

At $\alpha=0.05$, $df=1$:

$$\chi^2_{critical} = 3.841$$

Step 6: Decision

- Calculated $\chi^2 = 16.66$

$$\text{Critical } \chi^2 = 3.841$$

- Since $16.66 > 3.841$, we **reject H_0** .

Conclusion

At the 5% significance level, **product preference is not independent of gender.**

Hypothesis Testing

There is a significant association: males tend to prefer **Product A**, while females tend to prefer **Product B**.
